

✕ 1. Understanding Email Delivery

✕ Email Components:

Component	Function
MUA (Mail User Agent)	Client to read/send email (e.g., Thunderbird)
MTA (Mail Transfer Agent)	Sends/receives mail between servers (e.g., Postfix)
MDA (Mail Delivery Agent)	Delivers mail to user inbox (e.g., Dovecot)
Webmail	Access email via browser (e.g., SquirrelMail)

✕ How Mail is Delivered:

1.

User sends email using MUA ✕ submitted to Postfix (MTA).

2.

Postfix routes mail over the internet (SMTP protocol).

3.

Receiving server's Postfix accepts mail & passes it to **Dovecot (MDA)**.

4.

User retrieves mail via IMAP/POP3 (via Dovecot).

5.

Optional web interface via **SquirrelMail**.

❏ 2. Postfix Mail Server

❏ What is Postfix?

- Postfix is a **secure and fast MTA** that handles **SMTP-based email sending/receiving**.
- Installed in `/etc/postfix`.

☒ Installing Postfix:

On Debian/Ubuntu:

```
bash
```

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```
sudo apt install postfix
```

On RHEL/CentOS:

```
bash
```

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```
sudo yum install postfix
```

❏ Basic Configuration (/etc/postfix/main.cf):

conf

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```
myhostname = mail.example.com mydomain = example.com myorigin = /etc/mailname mydestination = $myhostname, localhost.$mydomain, localhost inet_interfaces = all
```

❏ Common Commands:

bash

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```
sudo systemctl start postfix sudo systemctl enable postfix mailq          # check mail queue postfix reload          # reload config
```

❏ 3. Dovecot – IMAP and POP3 Server

❏ What is Dovecot?

- Dovecot is an MDA and IMAP/POP3 server for delivering and allowing users to access their emails.

❏ Installing Dovecot:

bash

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```
sudo apt install dovecot-core dovecot-imapd dovecot-pop3d
```

❏ Key Configuration Files:

- /etc/dovecot/dovecot.conf

- `/etc/dovecot/conf.d/10-mail.conf`
- `/etc/dovecot/conf.d/10-auth.conf`

✂ Mail Location:

```
conf
```

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```
mail_location = maildir:~/Maildir
```

✂ Starting Dovecot:

```
bash
```

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```
sudo systemctl start dovecot sudo systemctl enable dovecot
```

❏ 4. SquirrelMail – Webmail Client

❏ What is SquirrelMail?

- SquirrelMail is a PHP-based webmail interface.
- Allows users to access their email using a browser.

❏ Installing:

Debian/Ubuntu:

```
bash
```

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```
sudo apt install squirrelmail
```

Configure:

```
bash
```

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```
sudo squirrelmail-configure
```

Access via:

```
arduino
```

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<http://localhost/squirrelmail>

Ensure Apache is running and PHP is installed.

☒ 5. Configuring Postfix + Dovecot + SquirrelMail (Integration)

1.

Set up Postfix as SMTP server.

2.

Configure Dovecot for mailbox access (Maildir, IMAP/POP3).

3.

Configure SquirrelMail to connect via **IMAP** to Dovecot.

4.

Create Linux users (email users):

```
bash
```

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```
sudo adduser testuser sudo passwd testuser
```

5.

Ensure Postfix delivers mail to /home/user/Maildir.